

Bellabeat Case Study

ASK — Define the Business Task

Bellabeat aims to better understand how consumers use smart devices in their daily lives. By analyzing publicly available Fitbit fitness tracker data, the goal is to identify user behavior trends and apply these insights to one Bellabeat product to support data-driven marketing strategy decisions.

Key Questions

1. What are the main trends in smart device usage?
2. How do these trends relate to Bellabeat's target customers?
3. How can these insights inform Bellabeat's marketing strategy?

Stakeholders

- Urška Sršen — Cofounder & Chief Creative Officer
- Sando Mur — Cofounder & Executive Team Member
- Bellabeat Marketing Analytics Team
- Bellabeat Executive Team

Business Value

Insights from this analysis will help Bellabeat:

- Understand how consumers interact with wellness devices
- Identify behaviors aligned with Bellabeat's target audience
- Strengthen product positioning and messaging
- Develop more effective marketing campaigns
- Support strategic growth in the smart device market

PREPARE — Describe the Data Sources

Data Source Overview

This analysis uses the Fitbit Fitness Tracker Dataset (CC0 Public Domain) available on Kaggle.

It includes daily activity, sleep, and heart rate data from 30 users.

Files Used

- `dailyActivity_merged.csv` — steps, calories, distance, activity levels
- `sleepDay_merged.csv` — minutes asleep, time in bed

These datasets provide sufficient information to identify daily behavior patterns.

Data Organization

Each file includes:

- Unique user ID
- Timestamps
- Activity or sleep metrics

Data Credibility (ROCCC Summary)

- **Reliable:** Medium — wearable sensor accuracy varies
- **Original:** Medium — real users, publicly shared
- **Comprehensive:** Medium — activity + sleep, but small sample
- **Current:** Low — data from 2016
- **Cited:** High — widely used in analytics education

Conclusion:

The dataset is appropriate for identifying general behavioral trends but not for large-scale predictions.

Data Limitations

- Small sample size (n=30)
- Users may not represent Bellabeat's demographic
- Data is several years old
- Sleep data incomplete for some users
- Heart rate data not consistently available

Despite limitations, the dataset is useful for exploring smart device usage patterns.

PROCESS — Clean and Prepare the Data

Tools Used

Python (Pandas, NumPy) was used for data cleaning, merging, and exploratory analysis.

Data Cleaning Steps

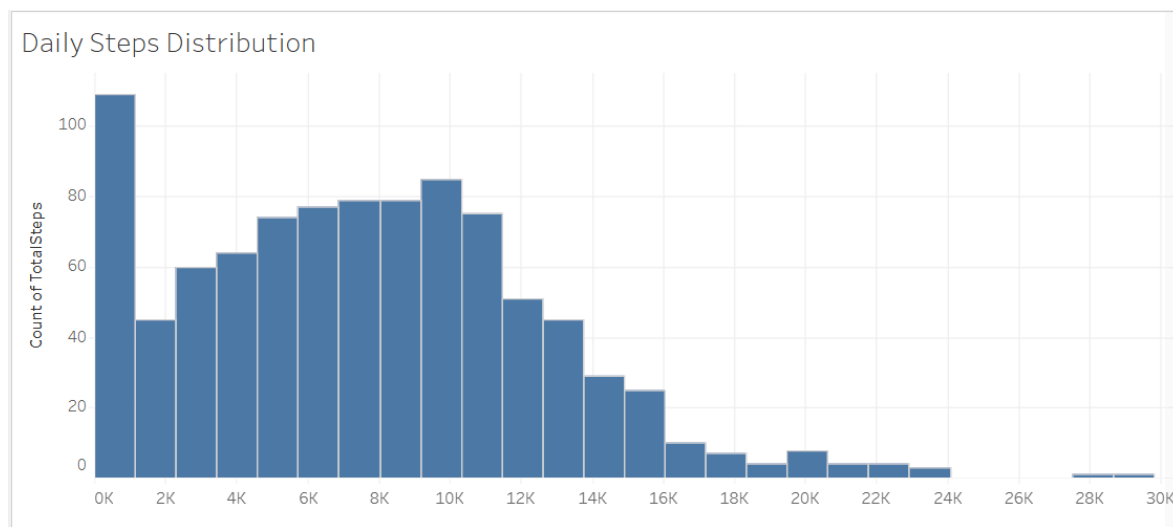
1. Removed duplicates in the sleep dataset
2. Standardized date formats across files
3. Renamed columns for consistency
4. Merged activity and sleep data on **Id** and **date**
5. Removed invalid entries (0 steps, 0 calories, 0 minutes asleep)
6. Created new calculated fields:
 - TotalActiveMinutes
 - SleepEfficiency
 - DayOfWeek
 - SedentaryPercentage

Validation

- No duplicates
- Valid timestamps
- Numeric fields within realistic ranges
- Merged dataset consistent

ANALYZE — Identify Trends and Generate Insights

Daily Steps Distribution



Insight:

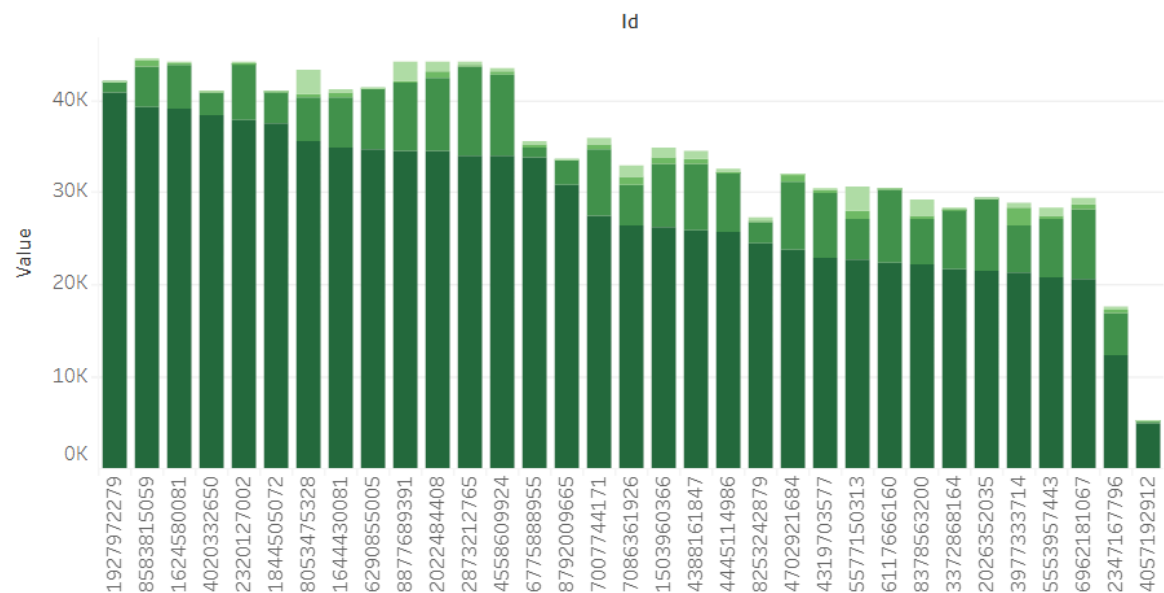
- Most users take 5,000–8,000 steps per day
- Few reach the recommended 10,000 steps
- Indicates generally low-to-moderate activity levels

Interpretation:

Bellabeat can encourage increased daily movement through reminders and challenges.

Sedentary vs Active Minutes

Sedentary vs Active Minutes



Insight:

- Users spend ~70–75% of their day sedentary
- Light activity dominates active minutes
- Very active minutes are minimal

Interpretation:

Bellabeat can support users with micro-movement prompts and routine-based nudges.

Descriptive Statistics

- Average daily steps: ~7,600
- Average sedentary minutes: ~990 (16.5 hours)
- Light activity is most common; high-intensity activity is rare

Sleep Analysis

- Average sleep: ~7 hours
- Sleep efficiency: ~91%
- Higher activity days correlate with slightly longer sleep

Weekly & Daily Patterns

- Weekdays: structured but sedentary
- Weekends: more variability
- Activity peaks between 5 PM–9 PM

User Segmentation

1. Sedentary Professionals
2. Lightly Active Users
3. Weekend Warriors
4. Highly Active Users

Key Trends

- High sedentary time
- Low–moderate activity
- Inconsistent sleep
- Predictable daily routines
- Different weekday/weekend behavior

SHARE — Communicate Findings

Key Insights

- Users spend most of the day sedentary
- Activity levels are generally low
- Sleep patterns vary widely
- Activity peaks in the evening
- Weekday vs weekend behavior differs

Key Messages for Stakeholders

- High sedentary time indicates need for movement guidance
- Low step counts show opportunity for activity-focused features
- Sleep inconsistency suggests value in sleep coaching
- Evening engagement is optimal for notifications
- Behavior patterns support personalized marketing

Recommended Visualizations

- Daily steps distribution
- Sedentary vs active minutes
- Sleep duration vs efficiency

- Hourly activity patterns
- Day-of-week activity trends
- User segmentation

Executive Summary

Smart device users show high sedentary time, low-to-moderate activity, and inconsistent sleep patterns. Their routines follow predictable cycles, with activity peaking in the evening. These insights highlight opportunities for Bellabeat to enhance wellness coaching, movement reminders, and sleep-focused features. Bellabeat can position its products as holistic wellness companions that support healthier daily habits.

ACT — Apply Insights and Recommend Next Steps

Top Recommendations

1. Position Bellabeat as a Holistic Wellness Coach

Emphasize combined insights on activity, sleep, stress, and routines.

2. Implement Personalized, Routine-Based Notifications

Examples:

- Afternoon movement prompts
- Evening wind-down routines
- Sleep preparation guidance

3. Create Segment-Specific Marketing Campaigns

- Sedentary Professionals → movement challenges
- Lightly Active Users → gentle routines
- Weekend Warriors → outdoor content
- Highly Active Users → performance insights

Next Steps for the Business

1. Integrate insights into upcoming marketing campaigns
2. Collaborate with product/UX teams to refine notifications
3. Expand data collection for personalization
4. Run A/B tests on wellness-focused messaging